

JUNGFRAUJOCH, Switzerland

WMO: 067300

Lat: 46.5475N

Lon: 7.9853E

Elev: 11732

StdP: 9.45

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-12.7	-9.4	-34.5	1.2	13.0	-29.5	1.6	13.3	56.7	5.9	50.6	7.6	16.5	310	1.156

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.9	45.1	35.6	42.9	34.6	40.9	33.6	38.9	42.5	37.3	40.5	35.8	38.8	7.6	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
36.9	50.3	39.5	35.3	47.2	38.0	33.9	44.6	36.8	17.7	42.7	16.8	40.7	16.0	39.0	46.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 49.1	42.2	35.2	DB	-15.7	50.6	4.1	2.9	-18.6	52.8	-21.0	54.5	-23.2	56.1	-26.2	58.3	(4)
(5)			WB	-16.2	42.3	3.8	2.0	-18.9	43.8	-21.2	45.0	-23.3	46.1	-26.0	47.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.3	8.9	8.2	11.5	15.6	23.2	30.1	33.1	33.7	27.9	24.1	15.3	11.1	
	DBStd	11.74	7.98	8.99	8.08	7.06	6.33	6.69	5.91	5.71	6.77	7.25	8.57	8.49	
	HDD50	10841	1273	1171	1193	1033	829	598	524	506	664	804	1041	1205	
	HDD65	16316	1738	1591	1658	1483	1294	1048	989	971	1114	1269	1491	1670	
	CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	
Wind	WSAvg	15.3	17.7	17.2	17.1	15.9	14.9	13.2	12.4	11.3	12.6	15.2	17.9	18.4	
Precipitation	PrecAvg	52.2	3.2	3.3	3.3	3.8	5.6	5.7	6.1	6.1	4.1	3.4	3.8	3.8	
	PrecMax	64.9	7.9	9.8	7.3	7.7	9.5	9.3	12.6	11.8	6.8	5.9	9.9	9.9	
	PrecMin	38.9	0.5	1.0	1.3	1.4	2.0	2.7	2.6	1.5	1.9	0.1	0.3	0.1	
	PrecStd	5.6	1.9	2.1	1.5	1.4	2.2	1.6	2.2	2.0	1.3	1.4	2.0	2.0	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.1	31.1	30.3	33.5	39.7	47.9	47.3	48.5	44.1	39.9	35.1	30.4	
		MCWB	19.9	20.6	22.4	24.1	31.8	35.5	38.6	36.7	31.7	28.0	23.9	22.4	
	2%	DB	24.8	26.7	26.4	29.9	36.2	43.7	44.2	45.1	40.8	36.8	31.4	25.8	
		MCWB	17.5	18.0	18.7	22.2	29.2	34.8	36.4	35.4	29.6	26.9	22.6	18.1	
	5%	DB	22.2	23.4	24.2	27.2	33.5	40.7	42.0	42.8	38.5	34.6	28.5	23.5	
		MCWB	16.9	16.8	17.8	21.7	28.0	33.7	35.1	34.7	29.0	26.0	21.3	17.3	
	10%	DB	19.6	20.3	21.7	24.4	31.0	38.4	40.0	40.7	36.3	32.7	25.9	21.5	
		MCWB	15.4	15.6	17.0	20.3	26.8	32.6	33.9	33.8	28.7	25.3	20.2	16.1	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.5	22.6	24.8	27.0	34.2	39.7	41.1	41.1	36.0	32.5	28.5	24.3
			MCDB	23.7	27.3	27.3	30.1	37.4	43.8	44.6	44.5	38.9	34.5	30.5	27.0
2%		WB	19.8	19.7	21.6	24.7	31.3	37.2	38.9	38.8	34.1	30.6	25.1	21.4	
		MCDB	22.4	23.6	23.9	27.6	34.0	40.7	42.2	42.5	37.0	33.0	27.6	23.0	
5%		WB	17.9	17.8	19.4	22.9	29.6	35.5	37.2	37.2	32.5	28.9	23.1	19.1	
		MCDB	21.0	22.3	22.1	25.6	32.0	38.8	40.4	40.4	35.1	31.7	26.4	21.3	
10%		WB	15.8	15.9	17.5	21.0	27.9	33.9	35.7	35.7	31.0	27.4	21.4	17.1	
		MCDB	19.2	20.4	20.6	23.5	30.2	37.1	38.6	38.6	33.5	30.6	24.8	19.8	
Mean Daily Temperature Range		5% DB	MDBR	9.5	9.4	9.2	9.1	8.8	8.9	8.3	7.9	8.0	8.1	8.3	9.2
			MCDBR	10.8	10.7	10.4	11.2	11.0	11.1	10.4	10.0	9.8	8.7	9.3	10.0
	MCWBR		8.7	8.1	8.3	9.3	9.3	9.9	9.4	9.1	8.4	6.7	7.2	8.6	
	5% WB	MCDBR	11.1	11.1	9.6	10.4	9.6	9.8	9.4	8.9	8.4	7.8	8.2	10.6	
		MCWBR	10.8	9.8	9.3	9.5	8.8	9.5	9.2	8.7	8.5	7.9	8.0	10.7	
Clear-Sky Solar Irradiance	taub	0.180	0.194	0.220	0.246	0.259	0.265	0.265	0.259	0.244	0.227	0.201	0.179		
	taud	2.446	2.417	2.412	2.459	2.519	2.539	2.528	2.564	2.607	2.635	2.579	2.469		
	Ebn at Noon	314	326	327	324	320	317	316	315	315	309	302	303		
	Edh at Noon	20	26	30	32	32	31	31	29	25	21	17	17		
All-Sky Solar Radiation	RadAvg	468	760	1131	1400	1549	1831	1814	1577	1234	809	475	386		
	RadStd	56	98	162	213	144	168	157	142	114	89	67	55		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.54	N/A	-3.35	+1.39	+0.90	+0.68	N/A	N/A	N/A	N/A
(47) Regional (118 neighbors)		N/A	N/A	N/A	+1.30	+0.76	+0.58	N/A	N/A	+103	+29

Nomenclature: See separate page